

6 (a) $F = BIL \sin \theta$ C1
 $= 2.6 \times 10^{-3} \times 5.4 \times 4.7 \times 10^{-2} \times \sin 34^\circ$
 $= 3.69 \times 10^{-4} \text{ N}$ A1 [2]
(allow 1 mark for use of $\cos 34^\circ$)

(b) peak current $= 1.7 \times \sqrt{2}$ C1
 $= 2.4 \text{ A}$

max. force $= 2.6 \times 10^{-3} \times 2.4 \times 4.7 \times 10^{-2} \times \sin 34^\circ$
 $= 1.64 \times 10^{-4} \text{ N}$ C1

variation $= 2 \times 1.64 \times 10^{-4}$
 $= 3.3 \times 10^{-4} \text{ N}$ A1 [3]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2014	9702	42